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SET08120 2018-9 TR1 001

Database Coursework

23/11/2018

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Section One - SELECT Statements (70 points)

11.Coincidence. Have two guests with the same surname ever stayed in the hotel at the same time? Show the last name and both first names. Do not include duplicates.

```
SELECT firstPerson.last_name,
firstPerson.first_name AS first_name,
secondPerson.first_name AS first_name
FROM
    (
        SELECT last_name,
        first_name,
        booking_date,
        nights
        FROM booking
        JOIN guest
        ON booking.guest_id = guest.id
    )as firstPerson,
    (
        SELECT last_name,
        first_name,
        booking_date,
        nights
        FROM booking
        JOIN guest
        ON booking.guest_id = guest.id
    )as secondPerson
    WHERE firstPerson.last_name = secondPerson.last_name AND
firstPerson.first_name != secondPerson.first_name
    AND
    (
        firstPerson.booking_date<=secondPerson.booking_date AND
firstPerson.booking_date + INTERVAL firstPerson.nights DAY >
secondPerson.booking_date
        OR
        secondPerson.booking_date<=firstPerson.booking_date AND
secondPerson.booking_date + INTERVAL secondPerson.nights DAY >
firstPerson.booking_date
    )
group by last_name
```

Result:

last_name	first_name	first_name
Davies	David T. C.	Philip
Evans	Mr Nigel	Graham
Howarth	Sir Gerald	Mr George
Jones	Susan Elan	Mr Marcus
Lewis	Dr Julian	Clive
McDonnell	Dr Alasdair	John

12. Check out per floor. The first digit of the room number indicates the floor – e.g. room 201 is on the 2nd floor. For each day of the week beginning 14th Nov 2016 show how many rooms are being vacated that day by floor number. Show all days in the correct order.

```
select leaving_date as i,
sum(case when floorNumber = 1 then checkouts end) as 1st,
sum(case when floorNumber = 2 then checkouts end) as 2nd,
sum(case when floorNumber = 3 then checkouts end) as 3rd
from
(
    SELECT leaving_date,
    floorNumber,
    COUNT(leaving_date) AS checkouts
    FROM
    (
        SELECT booking_date + INTERVAL nights DAY AS leaving_date,
        floor(room_no/100) AS floorNumber
        FROM booking
        WHERE booking_date + INTERVAL nights DAY BETWEEN '2016-11-14' AND
'2016-11-20'
    ) AS floorleavingdates
    GROUP BY leaving_date,floorNumber
) as amountleavingperfloor
group by leaving_date
```

Result:

i	1st	2nd	3rd
2016-11-14	5	3	4
2016-11-15	6	4	1
2016-11-16	2	2	4
2016-11-17	5	3	6
2016-11-18	2	3	2
2016-11-19	5	5	1
2016-11-20	2	2	2

13.Free rooms? List the rooms that are free on the day 25th Nov 2016.

```
SELECT id
FROM room
WHERE id NOT IN
(
    SELECT room_no
    FROM booking
    WHERE
      booking_date + INTERVAL nights DAY > '2016-11-25'
    AND
      booking_date <='2016-11-25'
)
ORDER BY id
```

Result:

id
207
210
304

14 Single room for three nights required. A customer wants a single room for three consecutive nights. Find the first available date in December 2016.

```
SELECT room_no as id,
MAX(leaving_date) as 'MIN(i)'
FROM
(
    SELECT room_no,
    booking_date+INTERVAL nights DAY AS leaving_date
    FROM booking
    JOIN room ON id = room_no
    WHERE room_type = 'single'
)roomsleavingdate
where leaving_date between '2016-12-1' and '2016-12-31'
GROUP BY room_no
ORDER BY MAX(leaving_date)
LIMIT 1
```

Result:

id	MIN(i)
201	2016-12-11

15 **Gross income by week.** Money is collected from guests when they leave. For each Thursday in November and December 2016 show the total amount of money collected from the previous Friday to that day inclusive.

```
SELECT AllDays AS Thursday,
COALESCE(weekly_income,0) AS weekly_income
FROM
(
    SELECT DATE('2017-2-02' - INTERVAL (firstvaluesselect.firstvalueset +
(10 * secondvalueselect.secondvalueset))*7 DAY) AS AllDays
    FROM
    (
        SELECT 0 AS firstvalueset
            UNION ALL SELECT 1
            UNION ALL SELECT 2
            UNION ALL SELECT 3
            UNION ALL SELECT 4
            UNION ALL SELECT 5
            UNION ALL SELECT 6
            UNION ALL SELECT 7
            UNION ALL SELECT 8
            UNION ALL SELECT 9
    ) AS firstvaluesselect
    CROSS JOIN
    (
        SELECT 0 AS secondvalueset
            UNION ALL SELECT 1
            UNION ALL SELECT 2
            UNION ALL SELECT 3
            UNION ALL SELECT 4
            UNION ALL SELECT 5
            UNION ALL SELECT 6
            UNION ALL SELECT 7
            UNION ALL SELECT 8
            UNION ALL SELECT 9
    ) AS secondvalueselect
)as daterange
LEFT JOIN
(
    SELECT ('2016-1-7'+ INTERVAL WEEK(booking.booking_date + INTERVAL
booking.nights - 5 DAY) WEEK) AS Thursday,
    SUM(booking.nights * rate.amount) + SUM(extra.amount) AS weekly_income
    FROM
    booking
    JOIN rate
    ON booking.occupants = rate.occupancy
    AND booking.room_type_requested = rate.room_type
    LEFT JOIN
    (
        SELECT booking_id,
        SUM(amount) AS amount
```

```
        FROM extra
        GROUP BY booking_id
    ) AS extra
    ON extra.booking_id = booking.booking_id
    GROUP BY Thursday
)as dateswithmoney
ON daterange.AllDays = dateswithmoney.Thursday
WHERE daterange.AllDays BETWEEN '2016-11-1' AND '2017-1-5'
ORDER BY Thursday
```

Result:

Thursday	weekly_income
2016-11-03	0.00
2016-11-10	12608.94
2016-11-17	13552.56
2016-11-24	12929.69
2016-12-01	11685.14
2016-12-08	13093.79
2016-12-15	8975.87
2016-12-22	1395.77
2016-12-29	0.00
2017-01-05	0.00

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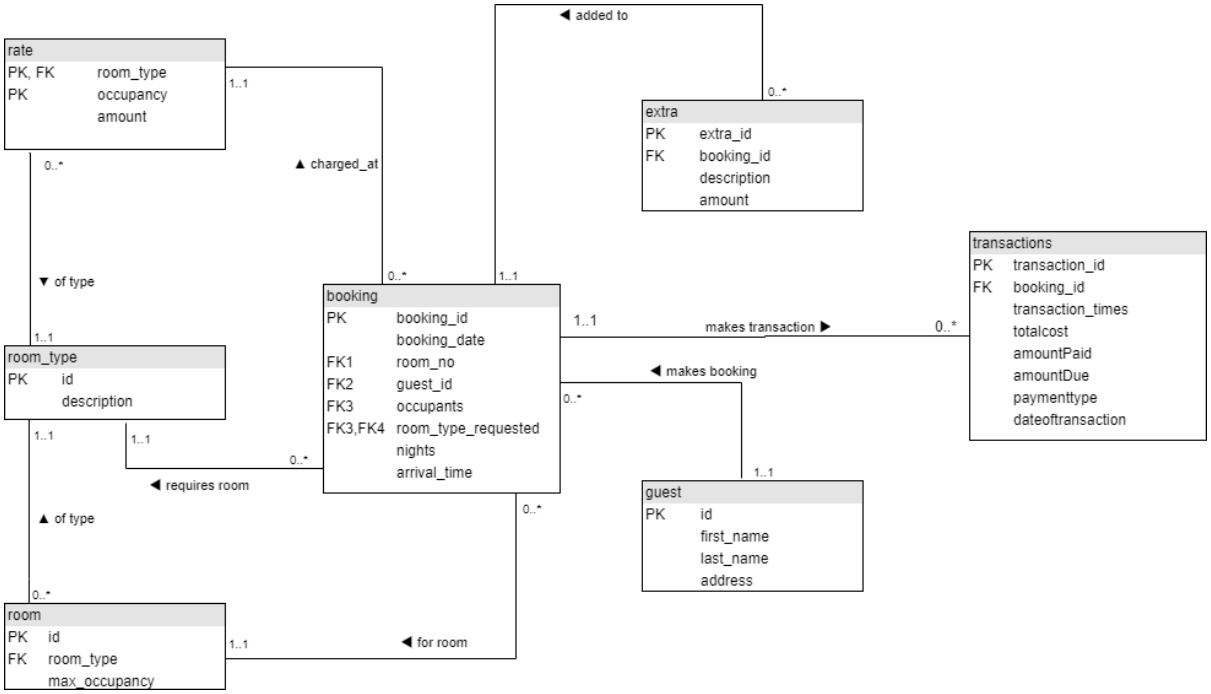
Section 2 – Database Design (30 points)

1)



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2a)

```
# this will create the inital table
create table transactions
(
    transaction_id int AUTO_INCREMENT,
    booking_id int,
    transaction_times DECIMAL(13, 2) default(0),
    totalcost DECIMAL(13, 2) default(0),
    amountPaid DECIMAL(13, 2)default(0),
    amount_due DECIMAL(13, 2) as (totalcost - amountPaid),
    paymenttype varchar(50),
    dateoftransaction DATE,
    primary key (transaction_id),
    foreign key (booking_id) references booking(booking_id)
);
```

2a)

```
#this will populate the table will all bookings
insert into transactions(totalcost, booking_id)
    select sum((booking.nights * rate.amount)+COALESCE(extra.amount,0)),booking.booking_id
    from booking
    join rate on
        booking.room_type_requested = rate.room_type and rate.occupancy = booking.occupants
        LEFT JOIN
                (
                SELECT booking_id,
                SUM(amount) AS amount
                FROM extra
                GROUP BY booking_id
                ) AS extra
        ON extra.booking_id = booking.booking_id
    group by booking.booking_id;
```

2b)

```
# For booking id 5360 the amount due (3 nights at £72 = £216) was paid by debit card when the guest checked out on
11th Dec
insert into transactions(booking_id,transaction_times,totalcost, amountPaid ,paymenttype ,dateoftransaction)
    select (booking_id),(transaction_times+1),(totalcost), (amountPaid+216) ,('debit') ,('2016-12-11')
```

```
    from transactions
    where booking_id = 5360
    order by transaction_times desc
    limit 1;
2b)

# For booking id 5359 the amount £288 was paid in advance by BACS (direct bank
# transfer) on 20th Nov
insert into transactions(booking_id,transaction_times,totalcost, amountPaid ,paymenttype ,dateoftransaction)
    select (booking_id),(transaction_times+1),(totalcost), (amountPaid+288) ,('BACS') ,('2016-11-20')
    from transactions
    where booking_id = 5359
    order by transaction_times desc
    limit 1;
2b)

#For the same booking 5359, the extras amount of £49.06 was paid in cash when the
#guest checked out
insert into transactions(booking_id,transaction_times,totalcost, amountPaid ,paymenttype ,dateoftransaction)
    select (booking_id),(transaction_times+1),(totalcost), (amountPaid+49.06) ,('cash') ,('2016-11-28')
    from transactions
    where booking_id = 5359
    order by transaction_times desc
    limit 1;
2b)

#The total bill for bookings 5012 and 5064 was settled at the end of November 2016
#with a single BACS payment.
insert into transactions(booking_id,transaction_times,totalcost, amountPaid ,paymenttype ,dateoftransaction)
    select (booking_id),(transaction_times+1),(totalcost), (totalcost) ,('BACS') ,('2016-11-30')
    from transactions
    where booking_id = 5012 or booking_id = 5064
    group by booking_id,transaction_times,totalcost, amountPaid ,paymenttype ,dateoftransaction
    order by transaction_times desc
    limit 2;
```